

Green Capital

Motivation, Research Question, and Literature

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GDP vs National Wealth

Gross Domestic Product (GDP)

- Measures economic activity over a period of time
- Flow variable (per year)

$$\text{GDP}_t = \sum_{i=1}^S \text{Value Added}_{i,t}$$

where:

- $\text{Value Added}_{i,t}$ is value added generated by sector i in period t ,
- S denotes the total number of economic sectors.

National Wealth

- Measures the stock of economic assets
- Determines future production and consumption possibilities

$$W_t = K_t + H_t + N_t$$

Key distinction:

GDP_t is a flow, W_t is a stock

National Wealth Decomposition

$$W_t = K_t + H_t + N_t$$

- K_t : Produced capital (infrastructure, machinery)
- H_t : Human capital (education, skills)
- N_t : Natural (green) capital

Natural (Green) Capital

- Renewable resources (forests, land, fisheries)
- Non-renewable resources (oil, gas, minerals)

Why Is This Important?

- Economic growth can occur alongside depletion of natural assets
- Climate change directly threatens natural capital stocks
- Loss of natural capital reduces future economic capacity
- GDP-based analysis fails to capture these long-run risks

Key implication:

Understanding national wealth is essential for assessing long-run sustainability.

Main Research Question

- How do different forms of capital contribute to national wealth?
- What role does natural capital play in wealth accumulation?
- Does this role differ across countries?

Focus of this study

- Wealth-based perspective rather than short-run GDP growth
- Comparison between a developed and an emerging economy
- Emphasis on long-run sustainability

Literature: Baseline Empirical Framework

Cobb–Douglas production framework

$$Y_t = A_t K_t^\alpha H_t^\beta N_t^\gamma \quad \Rightarrow \quad \ln Y_t = c + \alpha \ln K_t + \beta \ln H_t + \gamma \ln N_t$$

where:

- Y_t denotes output or national wealth at time t ,
- A_t is total factor productivity,
- K_t, H_t, N_t are types of capitals,
- α, β, γ are output elasticities.

Use in the literature:

- Most widely used framework in growth and wealth accounting
- Log-linear form allows direct interpretation of elasticities
- Provides a transparent benchmark for policy analysis

Growth-rate / dynamic models

$$\Delta \ln Y_{it} = \alpha \Delta \ln K_{it} + \beta \Delta \ln H_{it} + \gamma \Delta \ln N_{it} + u_{it}$$

where:

- $\Delta \ln Y_{it}$ is the growth rate of output or wealth,
- i indexes countries and t indexes time,
- u_{it} is an error term.

Use case:

- Removes common trends and mitigates multicollinearity
- Suitable for long panel datasets

Literature: Alternative Techniques II

CES production technology

$$Y_t = A_t (\alpha K_t^\rho + \beta H_t^\rho + \gamma N_t^\rho)^{1/\rho}, \quad \sigma = \frac{1}{1 - \rho}$$

where:

- Y_t is output or national wealth at time t ,
- A_t denotes total factor productivity,
- K_t , H_t , and N_t are produced, human, and natural capital,
- ρ governs the degree of substitutability between inputs,
- σ is the elasticity of substitution.

Interpretation: $\rho \rightarrow 0$: Cobb–Douglas; $\rho < 0$: complements; $\rho > 0$: substitutes.

Use case:

- Allows testing whether natural capital can be substituted by other capital
- Requires strong identification and rich data

This Study: Method and Cobb–Douglas

Why this method

- Short sample (1995–2020) and only two countries
- Focus on wealth stocks rather than high-frequency growth
- Need a simple, transparent specification

Chosen model

$$\ln W_t = c + \alpha \ln K_t + \beta \ln H_t + \gamma \ln N_t + \varepsilon_t$$

Why Cobb–Douglas

- Standard in growth and wealth accounting
- Coefficients are easy-to-interpret elasticities
- Fits the World Bank wealth data and sustainability focus

Data and Wealth Components

Data source

- World Bank: *Changing Wealth of Nations*
- Annual data, 1995–2020
- Countries: Germany and Brazil

National comprehensive wealth

$$W_t = K_t + H_t + R_t + N_t + (\text{other components})$$

Capital components used in the analysis

- K_t : Produced capital
- H_t : Human capital
- R_t : Renewable natural capital
- N_t : Non-renewable natural capital

All variables are expressed in real terms and used in logarithms where appropriate.

EDA & Country Comparison: Key KPIs

Research question: How do Germany and Brazil differ in the structure and dynamics of their national wealth and green capital (1995–2020)?

Human capital share

DEU

40.7%

BRA

48.7%

Renewable natural capital share

DEU

0.14%

BRA

2.83%

Wealth growth (national wealth)

DEU

1.67% / year

BRA

3.04% / year

Volatility of non-renewable NC

DEU

32.7%

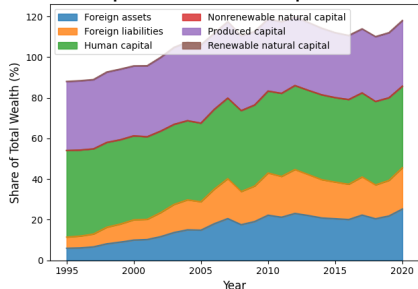
BRA

35.0%

Wealth Composition over Time

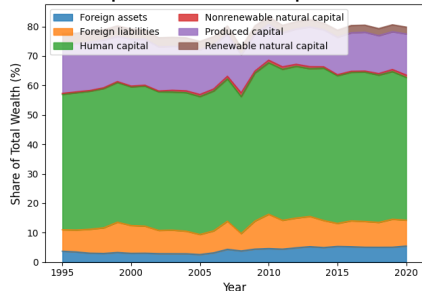
Germany (DEU)

DEU - Composition of National Comprehensive Wealth



Brazil (BRA)

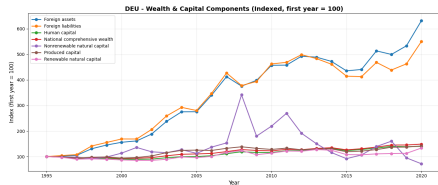
BRA - Composition of National Comprehensive Wealth



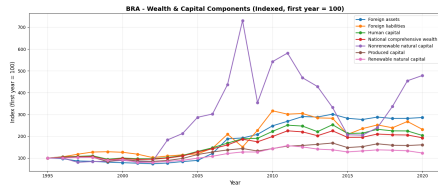
- **DEU:** wealth dominated by human ($\approx 41\%$) and produced capital ($\approx 35\%$); green capital $< 0.2\%$.
- **BRA:** human capital $\approx 49\%$, produced capital $\approx 14\%$; green capital clearly larger (renewable $\approx 2.8\%$, non-renewable $\approx 0.6\%$).

Dynamics: Growth, Volatility and Risk Profile

DEU – Indexed wealth and capital



BRA – Indexed wealth and capital



- Brazil's comprehensive wealth grows faster (3.0%) vs (1.7%) per year but with higher volatility (8.9%) vs (3.7%).
- Non-renewable natural capital growth is highly volatile in both ($\sigma \approx 33\text{--}35\%$).
- Renewable natural capital has much higher weight in Brazil, increasing exposure to **“green capital”** shocks.
- High correlation between capital types suggests potential multicollinearity, necessitating a parsimonious model

What EDA shows

- Wealth and capital stocks co-move strongly over time
- Human capital dominates wealth in both countries
- Natural capital is smaller in Germany but more important (and volatile) in Brazil

Implication

- Estimate a Cobb–Douglas model to quantify elasticities of wealth with respect to capital stocks
- Use a share-based specification as a robustness check for multicollinearity

Validity Check: Wealth Is Not a Mechanical Identity

Motivation

- National wealth is an accounting aggregate
- Capital stocks also appear as regressors
- Risk of a mechanical identity in regression analysis

Four-component capital sum

$$\text{SumCap}_t = K_t + H_t + R_t + N_t$$

Empirical verification

$$\text{Ratio}_t = \frac{W_t}{\text{SumCap}_t}$$

Key result

- National wealth is systematically larger than SumCap_t
- Average $\text{Ratio}_t \approx 1.43$
- Wealth is not a mechanical sum of the regressors

Cobb–Douglas Results: Germany vs Brazil

Estimated model

$$\ln(W_t) = \alpha + \beta_K \ln(K_t) + \beta_H \ln(H_t) + \beta_R \ln(R_t) + \beta_N \ln(N_t) + \varepsilon_t$$

Variable	Germany	Brazil
α (<i>constant</i>)	-1.3509	1.9792
Produced capital ($\ln K$)	0.378***	0.221***
Human capital ($\ln H$)	0.783***	0.734***
Renewable natural capital ($\ln R$)	-0.095**	-0.011 (n.s.)
Non-renewable natural capital ($\ln N$)	-0.010**	0.025***
R^2	0.998	0.999
Observations	26	26

Notes: *** $p < 0.01$, ** $p < 0.05$, n.s. not significant.

Main conclusions

- Human capital is the dominant driver of wealth in both countries
- Natural capital matters, but its role depends on development structure

Methodological insights

- Cobb–Douglas framework offers clarity and interpretability
- Multicollinearity is structural, not a statistical flaw

Stress testing

What if renewable capital drops 10 % ?

Implied wealth impact Germany: 0.95 %

Implied wealth impact Brazil: 0.11 %